

PCB connections — SV-16 Rev B

What this file is. The wiring of the SV-16 board, written out one part at a time and drawn, so that building the schematic or routing the board is a matter of copying rather than working things out. `PCB_COMPONENTS.md` says what each part is; this says what connects to what, in the order you build it.

Where it comes from. Every table below is generated (by `scripts/sv16_pcb_connections.py`) from the same data the KiCad board file is generated from — `scripts/sv16_board_kicad.py` for the netlist and `scripts/sv16_pcb_copper.py` for the copper plan. If a row here says `U1.54 -> R9 -> U2.6`, then the board file has that connection. Nothing in this document was typed from memory, and `make lint` fails if it drifts.

The schematic. `hardware/sv16_board/sv16_board.kicad_sch` is the same wiring as a KiCad schematic - 143 symbols, one per part, every pin labelled with its net (equal labels are the same net), drawn in the ten blocks below. It is generated from this netlist by `scripts/sv16_pcb_schematic.py`, so it cannot disagree with the board file; `make schematic` writes it, and `hardware/sv16_board/schematic_preview.png` is a picture of the whole sheet.

The pictures (in `hardware/sv16_board/`, regenerate with `make pcb-bitmap`):

File	What it shows
<code>pcb_top_view.png</code>	the board from above: every footprint, pad, silkscreen label, the In2 rail pours and all 266 vias
<code>pcb_bottom_view.png</code>	the same board flipped: through-hole pads and the B.Cu ground pour
<code>pcb_net_map.png</code>	the ratsnest coloured by function, with a legend — read this before you route
<code>pcb_power_map.png</code>	the power tree and the In2.Cu pour map, plus the pads the router still owes a trace
<code>pcb_connection_sheets.png</code>	the six connection sheets at the end of this file, drawn
<code>placement_preview.png</code>	the floorplan and the ratsnest, regenerated by the board generator
<code>schematic_preview.png</code>	the whole schematic sheet, drawn by <code>make schematic</code>

Two rules for reading it.

- Rows marked**  must be confirmed against the datasheet of the part in your hand (the MP1584's exposed pad, the USB-C and barrel-jack footprints, the crystal and inductor bodies). Everything else is standard-package or was reconciled against the cart in `hardware/cart/RECONCILED.md`.
- Net names are the LPF signal names** (`clk_25m`, `flash_cs_n`, `gpio_a[0]...`), so any net here can be traced to a line in `constraints/ecp5_144tqfp.lpf` and to a row in `board/TQFP144_PINOUT.md`. Power nets are `VM_IN` / `3V3` / `2V5` / `1V1` / `GND`.

1. Read this first — the five things that decide whether the board works

#	Thing	The rule	Where
1	The FPGA's rails	1V1 on pins 20/29/38/66/83/130, 2V5 on 17/53/96/132, 3V3 on 9/16/36/43/70/86/100/122/137. Getting one of these wrong is the classic dead-board bug	\$4
2	The Master-SPI pads are also GPIOA pins	gpio_a[1]/[2]/[4]/[6] share pins 46/47/49/51 with the configuration flash. Keep them inputs in firmware; R9-R12 (100 Ω) bound any contention	\$4, \$6
3	EN of the AP62300 is a 6 V pin	never tie it to VM_IN — the R48/R49 divider is what protects it	\$6
4	Y1 must be an active oscillator	the ECP5 has no crystal driver; a 2-pin HC49/US part leaves the board with no clock, no console, no boot	\$6
5	USB is a differential pair	J8 → U8 → U4, no vias, gap 0.2 mm, length skew under 0.5 mm, CC resistors 5.1 k to GND	\$4, \$8

2. Every net on the board

121 nets carry 472 pads. This is the whole list, with what each one is for; the pad-level detail is in §3.

net	pads	what it is
GND	103	the ground plane (In1.Cu) - every GND pad gets its own via
VM_IN_RAW	3	barrel / USB ORing node before FB1
VM_IN	8	7-12 V after the ferrite, feeds U5 and U6
3V3	63	main rail, AP62300 + L1; the whole In2.Cu pour
2V5	12	FPGA VCCAUX, LP5907 from 3V3, RC-delayed on EN
1V1	16	FPGA core, MP1584 + L2; patch pour on In2.Cu
5V_USB	3	VBUS only - J3/J4 pins 17/18/20, do not power the board here
CCLK	3	FPGA configuration clock (pin 54), R8 pull-up, R9 to U2
U2_CLK	2	U2 clock after the 100 ohm damping resistor
MOSI	2	FPGA configuration data out (pin 47)
U2_DI	2	U2 data in
MISO	2	FPGA configuration data in (pin 46)
U2_D0	2	U2 data out
CSSPIN	2	FPGA configuration chip select (pin 49)

net	pads	what it is
U2_CS	3	U2 chip select
U2_WP	2	U2 write protect (pulled high)
U2_HOLD	2	U2 hold (pulled high)
PROGRAMN	5	configuration trigger, active low
INITN	5	SRAM clear / config error
DONE	5	configuration complete, high when running
CFG_0	2	mode strap, low
CFG_1	2	mode strap, high (Master SPI)
CFG_2	2	mode strap, low
TCK	3	JTAG clock
TMS	3	JTAG mode select
TDI	3	JTAG data in
TD0	3	JTAG data out
ext_rst_n	4	reset button and RC, active low
Q1_G	3	Q1 gate (DTR auto-reconfigure)
DTR_F	2	DTR after the JP1 jumper
Q2_B	2	Q2 base
Q2_C	2	Q2 collector
Q4_B	2	Q4 base
Q4_C	2	Q4 collector
clk_25m	2	the 25 MHz clock into U1 pin 133
uart_rx	3	into the SoC (FPGA pin 73), idles high
uart_tx	3	out of the SoC (FPGA pin 74)
U4_TXD	2	CH340G transmit out
U4_RXD	2	CH340G receive in
flash_sck	2	application flash clock
flash_cs_n	3	application flash select
flash_mosi	2	application flash data out
flash_miso	2	application flash data in
U3_WP	2	U3 write protect (pulled high)
U3_HOLD	2	U3 hold (pulled high)
USB_VBUS	6	5 V from the USB-C connector
USB_DP	3	USB D+ at the connector
USB_DN	3	USB D- at the connector

net	pads	what it is
USB_DP_F	2	USB D+ after the ESD device
USB_DN_F	2	USB D- after the ESD device
CC1	2	USB-C CC1, 5.1 k to GND
CC2	2	USB-C CC2, 5.1 k to GND
U4_DTR	2	CH340G DTR# pin 13
U4_V3	2	CH340G V3 pin, tied to 3V3
Y2_XI	3	12 MHz crystal, XI side
Y2_X0	3	12 MHz crystal, XO side
U5_SW	3	AP62300 switch node
U5_BST	2	AP62300 bootstrap
U5_FB	3	AP62300 feedback node
U5_EN	3	AP62300 enable (6 V pin)
U6_SW	4	MP1584 switch node
U6_BST	2	MP1584 bootstrap
U6_EN	3	MP1584 enable RC
U6_FREQ	2	MP1584 frequency set (900 kHz)
FB_1V1	3	1V1 feedback node
U7_EN	3	LP5907 enable RC
pwm_out	2	motor PWM, 16 mA drive (pin 88)
motor_dir1	2	motor direction 1 (pin 89)
motor_dir2	2	motor direction 2 (pin 102)
motor_fault_n	3	motor fault input, pulled up (pin 103)
spi0_sck	2	SPI0 clock to EXP-A
spi0_cs_n	2	SPI0 chip select to EXP-A
spi0_mosi	2	SPI0 data out to EXP-A
spi0_miso	2	SPI0 data in to EXP-A
led[0]	2	
led[1]	2	
led[2]	2	
led[3]	2	
A0	2	
A1	2	
A2	2	
A3	2	
A4	2	

net	pads	what it is
A5	2	
A6	2	
A7	2	
A8	2	
A9	2	
A10	2	
A11	2	
A12	2	
A13	2	
A14	2	
A15	2	
B0	2	
B1	2	
B2	2	
B3	2	
B4	2	
B5	2	
B6	2	
B7	2	
B8	2	
B9	2	
B10	2	
B11	2	
B12	2	
B13	2	
B14	2	
B15	2	
D7_A	2	3V3-present LED anode
D1_A	2	D1 anode
D6_K	2	D6 cathode
J9_VIN	2	barrel jack positive pin
J10_TX	2	console header TX side
J10_RX	2	console header RX side
U6_COMP	3	MP1584 loop compensation
led[0]_k	2	

net	pads	what it is
led[1]_k	2	
led[2]_k	2	
led[3]_k	2	

3. Connection list, part by part

Read a row as: this pad, this net, and the other pads on that net. Power nets are collapsed (the GND plane (118 pads)) because they are the plane, not a wire — §7 says how each rail reaches its pads.

Build order

#	block	parts
1	Power tree	U5 · U6 · U7 · L1 · L2 · D8 · D9 · D11 · FB1 · C23 · C24 · C21 · C22 · C25 · C30 · C38 · C39 · C40 · C41 · C42 · R38 · R39 · R40 · R41 · R43 · R45 · R46 · R47 · R48 · R49 · C31 · C32 · C33 · C35 · J9
2	Clock, reset and status	Y1 · C26 · SW1 · SW2 · R1 · C1 · Q1 · Q2 · Q4 · R25 · R26 · R31 · R32 · R33 · R34 · R44 · D1 · D6 · D7 · JP1
3	Configuration flash	U2 · C28 · R5 · R6 · R7 · R8 · R9 · R10 · R11 · R12 · R13 · R14 · R15
4	Application flash	U3 · C29 · R16 · R17 · R18
5	USB and console	J8 · U8 · U4 · Y2 · C27 · C34 · C36 · C37 · R21 · R22 · R23 · R24 · R36 · R37 · J10
6	Motor and expansion	J3 · J4 · J5 · J6 · J7 · R19 · R27 · R28 · R29 · R30 · D2 · D3 · D4 · D5
7	JTAG	J1 · J2
8	The FPGA, its decoupling and the test points	U1 · C2 · C3 · C4 · C5 · C6 · C7 · C8 · C9 · C10 · C11 · C12 · C13 · C14 · C15 · C16 · C17 · C18 · C19 · C20 · TP1 · TP2 · TP3 · TP4 · TP5 · TP6 · H1 · H2 · H3 · H4 · FID1 · FID2 · FID3 · FID4

Power tree

U5 — AP62300TWU-7 (at 34.0, 11.0 mm)

C38 across BST(6)-SW(2) right at the pins; feedback tap at C40/C41, not at the far end

pad	net	also connected to
1	GND	the GND plane (102 pads)
2	U5_SW	L1.1, C38.2
3	VM_IN	VM_IN copper (7 pads)
4	U5_FB	R46.2, R47.1
5	U5_EN	R48.2, R49.1
6	U5_BST	C38.1

U6 — MP1584EN (at 26.0, 22.0 mm)

exposed pad soldered with its thermal vias; D11 loop as short as possible

pad	net	also connected to
1	U6_SW	D11.1, L2.1, C33.2
2	U6_EN	R41.2, C31.1
3	U6_COMP	R45.1, C35.1
4	FB_1V1	R38.2, R39.1
5	GND	the GND plane (102 pads)
6	U6_FREQ	R40.1
7	VM_IN	VM_IN copper (7 pads)
8	U6_BST	C33.1
9	GND	the GND plane (102 pads)

U7 — LP5907MFX-2.5 (at 38.0, 34.0 mm)

C42 must be low-ESR ceramic and close to OUT

pad	net	also connected to
1	3V3	the 3V3 pour (62 pads)
2	GND	the GND plane (102 pads)
3	U7_EN	R43.2, C32.1
5	2V5	the 2V5 patch (11 pads)

No net (leave unconnected): 4

L1 — 10 uH (at 42.0, 7.4 mm)

the AP62300 switch node: keep the loop tight

pad	net	also connected to
1	U5_SW	U5.2, C38.2
2	3V3	the 3V3 pour (62 pads)

L2 — 10 uH (at 31.7, 17.4 mm)

the MP1584 switch node: keep the loop tight

pad	net	also connected to
1	U6_SW	U6.1, D11.1, C33.2
2	1V1	the 1V1 patch (15 pads)

D8 — SS34 (at 18.0, 6.5 mm, rotated 90 deg)

pad	net	also connected to
1	VM_IN_RAW	the ORing node (2 pads)
2	J9_VIN	J9.1

D9 — SS34 (at 18.0, 13.4 mm, rotated 90 deg)

pad	net	also connected to
1	VM_IN_RAW	the ORing node (2 pads)
2	USB_VBUS	J8 VBUS (5 pads)

D11 — SS34 (at 32.0, 32.2 mm, rotated 90 deg)

pad	net	also connected to
1	U6_SW	U6.1, L2.1, C33.2
2	GND	the GND plane (102 pads)

FB1 — 600R @100MHz (at 22.0, 10.6 mm)

pad	net	also connected to
1	VM_IN_RAW	the ORing node (2 pads)
2	VM_IN	VM_IN copper (7 pads)

C23 — 470uF 25V (at 26.6, 5.1 mm)

input bulk, close to U5 VIN

pad	net	also connected to
1	VM_IN	VM_IN copper (7 pads)
2	GND	the GND plane (102 pads)

C24 — 220uF 16V (at 44.0, 20.4 mm)

3V3 output bulk, at the L1 side of the rail

pad	net	also connected to
1	3V3	the 3V3 pour (62 pads)
2	GND	the GND plane (102 pads)

C21 — 100uF 25V tant (at 30.0, 26.8 mm)

1V1 bulk, tantalum - check the polarity bar

pad	net	also connected to
1	1V1	the 1V1 patch (15 pads)
2	GND	the GND plane (102 pads)

C22 — 22uF 63V (at 40.0, 29.9 mm)

pad	net	also connected to
1	2V5	the 2V5 patch (11 pads)
2	GND	the GND plane (102 pads)

C25 — 10uF (at 44.0, 16.9 mm)

pad	net	also connected to
1	3V3	the 3V3 pour (62 pads)
2	GND	the GND plane (102 pads)

C30 — 10uF (at 60.0, 19.0 mm)

pad	net	also connected to
1	3V3	the 3V3 pour (62 pads)
2	GND	the GND plane (102 pads)

C38 — 100nF (at 37.8, 11.1 mm)

pad	net	also connected to
1	U5_BST	U5.6
2	U5_SW	U5.2, L1.1

C39 — 10uF (at 29.6, 8.6 mm)

pad	net	also connected to
1	VM_IN	VM_IN copper (7 pads)
2	GND	the GND plane (102 pads)

C40 — 22uF (at 46.0, 10.9 mm)

pad	net	also connected to
1	3V3	the 3V3 pour (62 pads)
2	GND	the GND plane (102 pads)

C41 — 22uF (at 46.0, 14.0 mm)

pad	net	also connected to
1	3V3	the 3V3 pour (62 pads)
2	GND	the GND plane (102 pads)

C42 — 10uF (at 43.0, 33.2 mm)

pad	net	also connected to
1	2V5	the 2V5 patch (11 pads)
2	GND	the GND plane (102 pads)

R38 — 12.4k (at 20.2, 23.7 mm)

pad	net	also connected to
1	1V1	the 1V1 patch (15 pads)
2	FB_1V1	U6.4, R39.1

R39 — 33k (at 21.0, 26.4 mm)

pad	net	also connected to
1	FB_1V1	U6.4, R38.2
2	GND	the GND plane (102 pads)

R40 — 100k (at 20.9, 29.1 mm)

pad	net	also connected to
1	U6_FREQ	U6.6
2	GND	the GND plane (102 pads)

R41 — 100k (at 24.7, 29.9 mm)

pad	net	also connected to
1	VM_IN	VM_IN copper (7 pads)
2	U6_EN	U6.2, C31.1

R43 — 100k (at 32.0, 39.5 mm)

pad	net	also connected to
1	3V3	the 3V3 pour (62 pads)
2	U7_EN	U7.3, C32.1

R45 — 10k (at 23.0, 18.0 mm)

pad	net	also connected to
1	U6_COMP	U6.3, C35.1
2	GND	the GND plane (102 pads)

R46 — 33k (at 39.3, 14.4 mm)

pad	net	also connected to
1	3V3	the 3V3 pour (62 pads)
2	U5_FB	U5.4, R47.1

R47 — 10k (at 33.9, 14.6 mm)

pad	net	also connected to
1	U5_FB	U5.4, R46.2
2	GND	the GND plane (102 pads)

R48 — 100k (at 27.0, 11.1 mm)

pad	net	also connected to
1	VM_IN	VM_IN copper (7 pads)
2	U5_EN	U5.5, R49.1

R49 — 33k (at 28.6, 13.8 mm)

pad	net	also connected to
1	U5_EN	U5.5, R48.2
2	GND	the GND plane (102 pads)

C31 — 470nF (at 28.4, 29.9 mm)

pad	net	also connected to
1	U6_EN	U6.2, R41.2
2	GND	the GND plane (102 pads)

C32 — 470nF (at 33.9, 37.0 mm)

pad	net	also connected to
1	U7_EN	U7.3, R43.2
2	GND	the GND plane (102 pads)

C33 — 100nF (at 22.0, 13.5 mm)

pad	net	also connected to
1	U6_BST	U6.8
2	U6_SW	U6.1, D11.1, L2.1

C35 — 100nF (at 23.0, 15.8 mm)

pad	net	also connected to
1	U6_COMP	U6.3, R45.1
2	GND	the GND plane (102 pads)

J9 — 7-12 V (at 10.0, 10.0 mm)

opening clear of the edge; keep the motor return away from it

pad	net	also connected to
1	J9_VIN	D8.2
2	GND	the GND plane (102 pads)

No net (leave unconnected): 3

Clock, reset and status

Y1 — 25 MHz ACTIVE XO (4-pin) (at 48.0, 32.0 mm)

within 10 mm of U1 pin 133, ground pour underneath, no via in the trace

pad	net	also connected to
1	3V3	the 3V3 pour (62 pads)
2	GND	the GND plane (102 pads)
3	clk_25m	U1.133
4	3V3	the 3V3 pour (62 pads)

C26 — 100nF (at 52.1, 34.0 mm)

pad	net	also connected to
1	3V3	the 3V3 pour (62 pads)
2	GND	the GND plane (102 pads)

SW1 — RESET (at 10.0, 62.0 mm)

pad	net	also connected to
1	ext_rst_n	U1.134, R1.1, C1.1
2	GND	the GND plane (102 pads)

No net (leave unconnected): 3, 4

SW2 — RECONFIG (at 20.0, 62.0 mm)

pad	net	also connected to
1	PROGRAMN	U1.57, Q1.3, R2.1, J1.4
2	GND	the GND plane (102 pads)

No net (leave unconnected): 3, 4

R1 — R_0603_1608Metric (at 5.6, 66.2 mm)

pad	net	also connected to
1	ext_rst_n	U1.134, C1.1, SW1.1
2	3V3	the 3V3 pour (62 pads)

C1 — 100nF (at 9.4, 66.9 mm)

pad	net	also connected to
1	ext_rst_n	U1.134, R1.1, SW1.1
2	GND	the GND plane (102 pads)

Q1 — 2N7002 (at 9.4, 70.0 mm)

pad	net	also connected to
1	Q1_G	R25.2, R26.1
2	GND	the GND plane (102 pads)
3	PROGRAMN	U1.57, R2.1, J1.4, SW2.1

Q2 — BC817 (at 28.8, 81.6 mm)

pad	net	also connected to
1	Q2_B	R32.2
2	GND	the GND plane (102 pads)
3	Q2_C	D1.1

Q4 — BC807 (at 3.3, 80.0 mm)

pad	net	also connected to
1	Q4_B	R31.2
2	3V3	the 3V3 pour (62 pads)
3	Q4_C	D6.2

R25 — R_0603_1608Metric (at 12.8, 70.0 mm)

pad	net	also connected to
1	DTR_F	JP1.2
2	Q1_G	Q1.1, R26.1

R26 — R_0603_1608Metric (at 10.0, 73.2 mm)

pad	net	also connected to
1	Q1_G	Q1.1, R25.2
2	GND	the GND plane (102 pads)

R31 — R_1206_3216Metric (at 8.1, 79.1 mm)

pad	net	also connected to
1	INITN	U1.55, R3.1, J1.10, TP6.1
2	Q4_B	Q4.1

R32 — R_0603_1608Metric (at 22.5, 72.5 mm)

pad	net	also connected to
1	DONE	U1.56, R4.1, J1.9, TP5.1
2	Q2_B	Q2.1

R33 — R_1206_3216Metric (at 33.5, 80.0 mm)

pad	net	also connected to
1	3V3	the 3V3 pour (62 pads)
2	D1_A	D1.2

R34 — R_1206_3216Metric (at 11.8, 82.2 mm)

pad	net	also connected to
1	D6_K	D6.1
2	GND	the GND plane (102 pads)

R44 — R_0603_1608Metric (at 39.4, 74.7 mm)

pad	net	also connected to
1	3V3	the 3V3 pour (62 pads)
2	D7_A	D7.2

D1 — LED green (at 28.7, 78.4 mm)

pad	net	also connected to
1	Q2_C	Q2.3
2	D1_A	R33.2

D6 — LED red (at 12.6, 78.7 mm) (not fitted)

pad	net	also connected to
1	D6_K	R34.1
2	Q4_C	Q4.3

D7 — LED red (at 35.6, 74.0 mm)

pad	net	also connected to
1	GND	the GND plane (102 pads)
2	D7_A	R44.2

JP1 — DTR ENABLE (at 6.0, 70.0 mm)

pad	net	also connected to
1	U4_DTR	U4.13
2	DTR_F	R25.1

Configuration flash

U2 — W25Q64JVSSIQ (at 60.0, 34.0 mm)

within 15 mm of U1 pins 54/46/47/49; R9-R12 at the FPGA end

pad	net	also connected to
1	U2_CS	R12.2, R13.1
2	U2_D0	R11.2
3	U2_WP	R14.1
4	GND	the GND plane (102 pads)
5	U2_DI	R10.2
6	U2_CLK	R9.2
7	U2_HOLD	R15.1
8	3V3	the 3V3 pour (62 pads)

C28 — 100nF (at 64.0, 38.0 mm)

pad	net	also connected to
1	3V3	the 3V3 pour (62 pads)
2	GND	the GND plane (102 pads)

R5 — R_0603_1608Metric (at 29.2, 70.0 mm)

pad	net	also connected to
1	CFG_1	U1.59
2	3V3	the 3V3 pour (62 pads)

R6 — R_0603_1608Metric (at 33.0, 70.0 mm)

pad	net	also connected to
1	CFG_0	U1.62

pad	net	also connected to
2	GND	the GND plane (102 pads)

R7 — R_0603_1608Metric (at 36.8, 70.0 mm)

pad	net	also connected to
1	CFG_2	U1.58
2	GND	the GND plane (102 pads)

R8 — R_0603_1608Metric (at 39.7, 72.4 mm)

pad	net	also connected to
1	CCLK	U1.54, R9.1
2	3V3	the 3V3 pour (62 pads)

R9 — R_1206_3216Metric (at 44.2, 73.2 mm)

100 ohm damping, at the FPGA end

pad	net	also connected to
1	CCLK	U1.54, R8.1
2	U2_CLK	U2.6

R10 — R_1206_3216Metric (at 44.9, 70.0 mm)

100 ohm damping, at the FPGA end

pad	net	also connected to
1	MOSI	U1.47
2	U2_DI	U2.5

R11 — R_1206_3216Metric (at 50.2, 71.2 mm)

100 ohm damping, at the FPGA end

pad	net	also connected to
1	MISO	U1.46
2	U2_D0	U2.2

R12 — R_1206_3216Metric (at 55.6, 72.0 mm)

100 ohm damping, at the FPGA end

pad	net	also connected to
1	CSSPIN	U1.49
2	U2_CS	U2.1, R13.1

R13 — R_0603_1608Metric (at 54.2, 30.0 mm)

pad	net	also connected to
1	U2_CS	U2.1, R12.2
2	3V3	the 3V3 pour (62 pads)

R14 — R_0603_1608Metric (at 58.0, 30.0 mm)

pad	net	also connected to
1	U2_WP	U2.3
2	3V3	the 3V3 pour (62 pads)

R15 — R_0603_1608Metric (at 61.8, 30.0 mm)

pad	net	also connected to
1	U2_HOLD	U2.7
2	3V3	the 3V3 pour (62 pads)

Application flash

U3 — W25Q64JVSSIQ (at 78.0, 52.0 mm)

ordinary SPI device on user I/O - nowhere near as critical as U2

pad	net	also connected to
1	flash_cs_n	U1.111, R16.1
2	flash_miso	U1.113
3	U3_WP	R17.1
4	GND	the GND plane (102 pads)
5	flash_mosi	U1.112
6	flash_sck	U1.110
7	U3_HOLD	R18.1
8	3V3	the 3V3 pour (62 pads)

C29 — 100nF (at 78.0, 57.0 mm)

pad	net	also connected to
1	3V3	the 3V3 pour (62 pads)
2	GND	the GND plane (102 pads)

R16 — R_0603_1608Metric (at 74.0, 45.8 mm)

pad	net	also connected to
1	flash_cs_n	U1.111, U3.1
2	3V3	the 3V3 pour (62 pads)

R17 — R_0603_1608Metric (at 74.0, 48.0 mm)

pad	net	also connected to
1	U3_WP	U3.3
2	3V3	the 3V3 pour (62 pads)

R18 — R_0603_1608Metric (at 74.0, 56.0 mm)

pad	net	also connected to
1	U3_HOLD	U3.7
2	3V3	the 3V3 pour (62 pads)

USB and console

J8 — USB-C (at 90.0, 88.0 mm)

flush with the board edge, CC resistors within 10 mm

pad	net	also connected to
A1	GND	the GND plane (102 pads)
A4	USB_VBUS	J8 VBUS (5 pads)
A5	CC1	R36.1
A6	USB_DP	U8.1, J8.B6
A7	USB_DN	U8.3, J8.B7
A9	USB_VBUS	J8 VBUS (5 pads)
B1	GND	the GND plane (102 pads)
B4	USB_VBUS	J8 VBUS (5 pads)
B5	CC2	R37.1
B6	USB_DP	U8.1, J8.A6
B7	USB_DN	U8.3, J8.A7
B9	USB_VBUS	J8 VBUS (5 pads)
S1	GND	the GND plane (102 pads)
S2	GND	the GND plane (102 pads)
S3	GND	the GND plane (102 pads)
S4	GND	the GND plane (102 pads)
A12	GND	the GND plane (102 pads)
B12	GND	the GND plane (102 pads)

No net (leave unconnected): A8, B8

U8 — USBLC6-2SC6 (at 83.0, 88.0 mm)

at the connector, not at the chip - it protects U4

pad	net	also connected to
1	USB_DP	J8.A6, J8.B6
2	GND	the GND plane (102 pads)
3	USB_DN	J8.A7, J8.B7
4	USB_DN_F	U4.6
5	USB_VBUS	J8 VBUS (5 pads)
6	USB_DP_F	U4.5

U4 — CH340G (at 74.0, 84.0 mm)

next to J8; keep XI/XO short and symmetrical, away from the switchers

pad	net	also connected to
1	GND	the GND plane (102 pads)
2	U4_TXD	R21.1
3	U4_RXD	R23.1
4	U4_V3	C34.1
5	USB_DP_F	U8.6
6	USB_DN_F	U8.4
7	Y2_XI	Y2.1, C36.1
8	Y2_XO	Y2.2, C37.1
13	U4_DTR	JP1.1
16	3V3	the 3V3 pour (62 pads)

No net (leave unconnected): 9, 10, 11, 12, 14, 15

Y2 — 12 MHz (CH340G clock) (at 70.0, 92.0 mm)

short, symmetrical XI/XO traces; load caps to the same ground

pad	net	also connected to
1	Y2_XI	U4.7, C36.1
2	Y2_XO	U4.8, C37.1

C27 — 100nF (at 70.0, 89.8 mm)

pad	net	also connected to
1	3V3	the 3V3 pour (62 pads)
2	GND	the GND plane (102 pads)

C34 — 100nF (at 69.8, 77.5 mm)

pad	net	also connected to
1	U4_V3	U4.4
2	GND	the GND plane (102 pads)

C36 — 100nF (at 66.2, 89.5 mm)

pad	net	also connected to
1	Y2_XI	U4.7, Y2.1
2	GND	the GND plane (102 pads)

C37 — 100nF (at 73.8, 90.2 mm)

pad	net	also connected to
1	Y2_XO	U4.8, Y2.2
2	GND	the GND plane (102 pads)

R21 — R_0603_1608Metric (at 69.6, 75.2 mm)

pad	net	also connected to
1	U4_TXD	U4.2
2	uart_rx	U1.73, R22.1

R22 — R_0603_1608Metric (at 65.6, 76.0 mm)

pad	net	also connected to
1	uart_rx	U1.73, R21.2
2	J10_RX	J10.3

R23 — R_0603_1608Metric (at 73.6, 77.4 mm)

pad	net	also connected to
1	U4_RXD	U4.3
2	uart_tx	U1.74, R24.1

R24 — R_0603_1608Metric (at 69.4, 73.0 mm)

pad	net	also connected to
1	uart_tx	U1.74, R23.2
2	J10_TX	J10.2

R36 — 5.1k (at 85.0, 93.8 mm)

pad	net	also connected to
1	CC1	J8.A5
2	GND	the GND plane (102 pads)

R37 — 5.1k (at 89.2, 93.8 mm)

pad	net	also connected to
1	CC2	J8.B5
2	GND	the GND plane (102 pads)

J10 — CONSOLE (at 62.0, 88.0 mm)

pad	net	also connected to
1	3V3	the 3V3 pour (62 pads)
2	J10_TX	R24.2
3	J10_RX	R22.2
4	GND	the GND plane (102 pads)

Motor and expansion

J3 — EXP-A (at 52.0, 86.5 mm)

pad	net	also connected to
1	3V3	the 3V3 pour (62 pads)
2	GND	the GND plane (102 pads)
3	A8	U1.97
4	A9	U1.98
5	A10	U1.99
6	A11	U1.104
7	A12	U1.105
8	A13	U1.106
9	A14	U1.107
10	A15	U1.108
11	spi0_cs_n	U1.115
12	spi0_sck	U1.114
13	spi0_mosi	U1.116
14	spi0_miso	U1.117
15	GND	the GND plane (102 pads)
16	GND	the GND plane (102 pads)
17	5V_USB	J3/J4 5 V pins (2 pads)
18	5V_USB	J3/J4 5 V pins (2 pads)
19	GND	the GND plane (102 pads)
20	GND	the GND plane (102 pads)

J4 — EXP-B (at 24.0, 86.5 mm)

pad	net	also connected to
1	3V3	the 3V3 pour (62 pads)
2	GND	the GND plane (102 pads)
3	B0	U1.135
4	B1	U1.136
5	B2	U1.139
6	B3	U1.140
7	B4	U1.141
8	B5	U1.142
9	B6	U1.143
10	B7	U1.128
11	B8	U1.124
12	B9	U1.125
13	B10	U1.126

pad	net	also connected to
14	B11	U1.127
15	B12	U1.1
16	B13	U1.2
17	B14	U1.3
18	B15	U1.4
19	GND	the GND plane (102 pads)
20	5V_USB	J3/J4 5 V pins (2 pads)

J5 — EXP-C (at 18.0, 86.5 mm)

pad	net	also connected to
1	A0	U1.45
2	A1	U1.46
3	A2	U1.47
4	A3	U1.48
5	A4	U1.49
6	A5	U1.50
7	A6	U1.51
8	A7	U1.52
9	GND	the GND plane (102 pads)
10	3V3	the 3V3 pour (62 pads)

J6 — MOTOR (at 86.0, 70.0 mm)

pad	net	also connected to
1	VM_IN	VM_IN copper (7 pads)
2	GND	the GND plane (102 pads)
3	pwm_out	U1.88
4	motor_dir1	U1.89
5	motor_dir2	U1.102
6	motor_fault_n	U1.103, R19.1

J7 — SPARE (not fitted) (at 93.0, 50.0 mm) (not fitted)

pad	net	also connected to

No net (leave unconnected): 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50

R19 — R_0603_1608Metric (at 82.0, 73.0 mm)

pad	net	also connected to
1	motor_fault_n	U1.103, J6.6

pad	net	also connected to
2	3V3	the 3V3 pour (62 pads)

R27 — R_0805_2012Metric (at 34.5, 85.0 mm)

pad	net	also connected to
1	led[0]	U1.39
2	led[0]_k	D2.1

R28 — R_0805_2012Metric (at 38.7, 85.0 mm)

pad	net	also connected to
1	led[1]	U1.40
2	led[1]_k	D3.1

R29 — R_0805_2012Metric (at 42.9, 85.0 mm)

pad	net	also connected to
1	led[2]	U1.41
2	led[2]_k	D4.1

R30 — R_0805_2012Metric (at 47.1, 85.0 mm)

pad	net	also connected to
1	led[3]	U1.44
2	led[3]_k	D5.1

D2 — LED green (at 36.0, 88.0 mm)

pad	net	also connected to
1	led[0]_k	R27.2
2	3V3	the 3V3 pour (62 pads)

D3 — LED green (at 39.8, 88.0 mm)

pad	net	also connected to
1	led[1]_k	R28.2
2	3V3	the 3V3 pour (62 pads)

D4 — LED green (at 43.6, 88.0 mm)

pad	net	also connected to
1	led[2]_k	R29.2
2	3V3	the 3V3 pour (62 pads)

D5 — LED green (at 47.3, 88.0 mm)

pad	net	also connected to
1	led[3]_k	R30.2

pad	net	also connected to
2	3V3	the 3V3 pour (62 pads)

JTAG

J1 — JTAG (at 16.0, 78.0 mm)

pad	net	also connected to
1	3V3	the 3V3 pour (62 pads)
2	TD0	U1.60, J2.4
3	TDI	U1.61, J2.3
4	PROGRAMN	U1.57, Q1.3, R2.1, SW2.1
6	TMS	U1.64, J2.2
7	GND	the GND plane (102 pads)
8	TCK	U1.63, J2.5
9	DONE	U1.56, R4.1, R32.1, TP5.1
10	INITN	U1.55, R3.1, R31.1, TP6.1

No net (leave unconnected): 5

J2 — JTAG-1.27 (at 10.0, 88.0 mm)

pad	net	also connected to
1	3V3	the 3V3 pour (62 pads)
2	TMS	U1.64, J1.6
3	TDI	U1.61, J1.3
4	TD0	U1.60, J1.2
5	TCK	U1.63, J1.8
6	GND	the GND plane (102 pads)
10	GND	the GND plane (102 pads)

No net (leave unconnected): 7, 8, 9

The FPGA, its decoupling and the test points

U1 — LFE5U-12F-6TG144C (at 52.0, 52.0 mm)

0.5 mm pitch: 0.2 mm fan-out tracks, 0.45/0.2 mm vias; fit it LAST, after the rails measure right

pad	net	also connected to
1	B12	J4.15
2	B13	J4.16
3	B14	J4.17
4	B15	J4.18
8	GND	the GND plane (102 pads)

pad	net	also connected to
9	3V3	the 3V3 pour (62 pads)
15	GND	the GND plane (102 pads)
16	3V3	the 3V3 pour (62 pads)
17	2V5	the 2V5 patch (11 pads)
20	1V1	the 1V1 patch (15 pads)
21	GND	the GND plane (102 pads)
29	1V1	the 1V1 patch (15 pads)
32	GND	the GND plane (102 pads)
36	3V3	the 3V3 pour (62 pads)
38	1V1	the 1V1 patch (15 pads)
39	led[0]	R27.1
40	led[1]	R28.1
41	led[2]	R29.1
42	GND	the GND plane (102 pads)
43	3V3	the 3V3 pour (62 pads)
44	led[3]	R30.1
45	A0	J5.1
46	A1	J5.2
47	A2	J5.3
48	A3	J5.4
49	A4	J5.5
50	A5	J5.6
51	A6	J5.7
52	A7	J5.8
53	2V5	the 2V5 patch (11 pads)
54	CCLK	R8.1, R9.1
55	INITN	R3.1, R31.1, J1.10, TP6.1
56	DONE	R4.1, R32.1, J1.9, TP5.1
57	PROGRAMN	Q1.3, R2.1, J1.4, SW2.1
58	CFG_2	R7.1
59	CFG_1	R5.1
60	TD0	J1.2, J2.4
61	TDI	J1.3, J2.3
62	CFG_0	R6.1
63	TCK	J1.8, J2.5

pad	net	also connected to
64	TMS	J1.6, J2.2
65	GND	the GND plane (102 pads)
66	1V1	the 1V1 patch (15 pads)
70	3V3	the 3V3 pour (62 pads)
73	uart_rx	R21.2, R22.1
74	uart_tx	R23.2, R24.1
75	GND	the GND plane (102 pads)
83	1V1	the 1V1 patch (15 pads)
85	GND	the GND plane (102 pads)
86	3V3	the 3V3 pour (62 pads)
87	GND	the GND plane (102 pads)
88	pwm_out	J6.3
89	motor_dir1	J6.4
96	2V5	the 2V5 patch (11 pads)
97	A8	J3.3
98	A9	J3.4
99	A10	J3.5
100	3V3	the 3V3 pour (62 pads)
101	GND	the GND plane (102 pads)
102	motor_dir2	J6.5
103	motor_fault_n	R19.1, J6.6
104	A11	J3.6
105	A12	J3.7
106	A13	J3.8
107	A14	J3.9
108	A15	J3.10
110	flash_sck	U3.6
111	flash_cs_n	U3.1, R16.1
112	flash_mosi	U3.5
113	flash_miso	U3.2
114	spi0_sck	J3.12
115	spi0_cs_n	J3.11
116	spi0_mosi	J3.13
117	spi0_miso	J3.14
122	3V3	the 3V3 pour (62 pads)

pad	net	also connected to
123	GND	the GND plane (102 pads)
124	B8	J4.11
125	B9	J4.12
126	B10	J4.13
127	B11	J4.14
128	B7	J4.10
129	GND	the GND plane (102 pads)
130	1V1	the 1V1 patch (15 pads)
131	GND	the GND plane (102 pads)
132	2V5	the 2V5 patch (11 pads)
133	clk_25m	Y1.3
134	ext_rst_n	R1.1, C1.1, SW1.1
135	B0	J4.3
136	B1	J4.4
137	3V3	the 3V3 pour (62 pads)
138	GND	the GND plane (102 pads)
139	B2	J4.5
140	B3	J4.6
141	B4	J4.7
142	B5	J4.8
143	B6	J4.9

No net (leave unconnected): 5, 6, 7, 10, 11, 12, 13, 14, 18, 19, 22, 23, 24, 25, 26, 27, 28, 30, 31, 33, 34, 35, 37, 67, 68, 69, 71, 72, 76, 77, 78, 79, 80, 81, 82, 84, 90, 91, 92, 93, 94, 95, 109, 118, 119, 120, 121, 144

C2 — 100nF (at 34.0, 46.0 mm)

pad	net	also connected to
1	1V1	the 1V1 patch (15 pads)
2	GND	the GND plane (102 pads)

C3 — 100nF (at 34.0, 49.0 mm)

pad	net	also connected to
1	1V1	the 1V1 patch (15 pads)
2	GND	the GND plane (102 pads)

C4 — 100nF (at 34.0, 52.0 mm)

pad	net	also connected to
1	1V1	the 1V1 patch (15 pads)

pad	net	also connected to
2	GND	the GND plane (102 pads)

C5 — 100nF (at 34.0, 55.0 mm)

pad	net	also connected to
1	1V1	the 1V1 patch (15 pads)
2	GND	the GND plane (102 pads)

C6 — 100nF (at 34.0, 58.0 mm)

pad	net	also connected to
1	1V1	the 1V1 patch (15 pads)
2	GND	the GND plane (102 pads)

C7 — 100nF (at 36.5, 61.0 mm, rotated 90 deg)

pad	net	also connected to
1	1V1	the 1V1 patch (15 pads)
2	GND	the GND plane (102 pads)

C8 — 100nF (at 42.4, 37.5 mm)

pad	net	also connected to
1	2V5	the 2V5 patch (11 pads)
2	GND	the GND plane (102 pads)

C9 — 100nF (at 46.2, 37.5 mm)

pad	net	also connected to
1	2V5	the 2V5 patch (11 pads)
2	GND	the GND plane (102 pads)

C10 — 100nF (at 50.0, 37.5 mm)

pad	net	also connected to
1	2V5	the 2V5 patch (11 pads)
2	GND	the GND plane (102 pads)

C11 — 100nF (at 53.8, 37.5 mm)

pad	net	also connected to
1	2V5	the 2V5 patch (11 pads)
2	GND	the GND plane (102 pads)

C12 — 100nF (at 67.0, 46.0 mm)

pad	net	also connected to
1	3V3	the 3V3 pour (62 pads)

pad	net	also connected to
2	GND	the GND plane (102 pads)

C13 — 100nF (at 67.0, 49.0 mm)

pad	net	also connected to
1	3V3	the 3V3 pour (62 pads)
2	GND	the GND plane (102 pads)

C14 — 100nF (at 67.0, 52.0 mm)

pad	net	also connected to
1	3V3	the 3V3 pour (62 pads)
2	GND	the GND plane (102 pads)

C15 — 100nF (at 67.0, 55.0 mm)

pad	net	also connected to
1	3V3	the 3V3 pour (62 pads)
2	GND	the GND plane (102 pads)

C16 — 100nF (at 67.0, 58.0 mm)

pad	net	also connected to
1	3V3	the 3V3 pour (62 pads)
2	GND	the GND plane (102 pads)

C17 — 100nF (at 69.2, 61.0 mm, rotated 90 deg)

pad	net	also connected to
1	3V3	the 3V3 pour (62 pads)
2	GND	the GND plane (102 pads)

C18 — 100nF (at 65.1, 61.0 mm, rotated 90 deg)

pad	net	also connected to
1	3V3	the 3V3 pour (62 pads)
2	GND	the GND plane (102 pads)

C19 — 100nF (at 67.0, 61.0 mm, rotated 90 deg)

pad	net	also connected to
1	3V3	the 3V3 pour (62 pads)
2	GND	the GND plane (102 pads)

C20 — 100nF (at 38.7, 61.0 mm, rotated 90 deg)

pad	net	also connected to
1	3V3	the 3V3 pour (62 pads)

pad	net	also connected to
2	GND	the GND plane (102 pads)

TP1 — 3V3 (at 46.0, 4.3 mm)

pad	net	also connected to
1	3V3	the 3V3 pour (62 pads)

TP2 — 2V5 (at 41.5, 26.9 mm)

pad	net	also connected to
1	2V5	the 2V5 patch (11 pads)

TP3 — 1V1 (at 22.0, 32.1 mm)

pad	net	also connected to
1	1V1	the 1V1 patch (15 pads)

TP4 — GND (at 55.0, 6.0 mm)

pad	net	also connected to
1	GND	the GND plane (102 pads)

TP5 — DONE (at 45.8, 75.8 mm)

pad	net	also connected to
1	DONE	U1.56, R4.1, R32.1, J1.9

TP6 — INITN (at 48.1, 75.7 mm)

pad	net	also connected to
1	INITN	U1.55, R3.1, R31.1, J1.10

H1 — MountingHole_3.2mm_M3 (at 4.0, 4.0 mm)

pad	net	also connected to

H2 — MountingHole_3.2mm_M3 (at 96.0, 4.0 mm)

pad	net	also connected to

H3 — MountingHole_3.2mm_M3 (at 4.0, 96.0 mm)

pad	net	also connected to

H4 — MountingHole_3.2mm_M3 (at 96.0, 96.0 mm)

pad	net	also connected to

FID1 — Fiducial_1mm_Mask2mm (at 8.0, 20.0 mm)

pad	net	also connected to

No net (leave unconnected): 1

FID2 — Fiducial_1mm_Mask2mm (at 92.0, 14.0 mm)

pad	net	also connected to

No net (leave unconnected): 1

FID3 — Fiducial_1mm_Mask2mm (at 8.0, 36.0 mm)

pad	net	also connected to

No net (leave unconnected): 1

FID4 — Fiducial_1mm_Mask2mm (at 92.5, 93.8 mm)

pad	net	also connected to

No pad (leave unconnected): 1

4. The FPGA — U1, LFE5U-12F-6TG144C

52 of the 98 I/O pins are used. Pin numbers are the TQFP-144 physical pins; the signal names are the LPF names, so each row can be traced to `constraints/ecp5_144tqfp.lpf`.

TQFP pin	signal	board net	connects to
1	gpio_b[12]	B12	J4.15
2	gpio_b[13]	B13	J4.16
3	gpio_b[14]	B14	J4.17
4	gpio_b[15]	B15	J4.18
39	led[0]	led[0]	R27.1
40	led[1]	led[1]	R28.1
41	led[2]	led[2]	R29.1
44	led[3]	led[3]	R30.1
45	gpio_a[0]	A0	J5.1
46	gpio_a[1]	A1	J5.2
47	gpio_a[2]	A2	J5.3
48	gpio_a[3]	A3	J5.4
49	gpio_a[4]	A4	J5.5
50	gpio_a[5]	A5	J5.6
51	gpio_a[6]	A6	J5.7
52	gpio_a[7]	A7	J5.8
73	uart_rx	uart_rx	R21.2, R22.1
74	uart_tx	uart_tx	R23.2, R24.1
88	pwm_out	pwm_out	J6.3
89	motor_dir1	motor_dir1	J6.4

TQFP pin	signal	board net	connects to
97	gpio_a[8]	A8	J3.3
98	gpio_a[9]	A9	J3.4
99	gpio_a[10]	A10	J3.5
102	motor_dir2	motor_dir2	J6.5
103	motor_fault_n	motor_fault_n	R19.1, J6.6
104	gpio_a[11]	A11	J3.6
105	gpio_a[12]	A12	J3.7
106	gpio_a[13]	A13	J3.8
107	gpio_a[14]	A14	J3.9
108	gpio_a[15]	A15	J3.10
110	flash_sck	flash_sck	U3.6
111	flash_cs_n	flash_cs_n	U3.1, R16.1
112	flash_mosi	flash_mosi	U3.5
113	flash_miso	flash_miso	U3.2
114	spi0_sck	spi0_sck	J3.12
115	spi0_cs_n	spi0_cs_n	J3.11
116	spi0_mosi	spi0_mosi	J3.13
117	spi0_miso	spi0_miso	J3.14
124	gpio_b[8]	B8	J4.11
125	gpio_b[9]	B9	J4.12
126	gpio_b[10]	B10	J4.13
127	gpio_b[11]	B11	J4.14
128	gpio_b[7]	B7	J4.10
133	clk_25m	clk_25m	Y1.3
134	ext_rst_n	ext_rst_n	R1.1, C1.1, SW1.1
135	gpio_b[0]	B0	J4.3
136	gpio_b[1]	B1	J4.4
139	gpio_b[2]	B2	J4.5
140	gpio_b[3]	B3	J4.6
141	gpio_b[4]	B4	J4.7
142	gpio_b[5]	B5	J4.8
143	gpio_b[6]	B6	J4.9

Supplies and ground (the rows that matter most):

pins	rail	decoupling
20, 29, 38, 66, 83, 130	1V1	C2-C7, 100 nF each

pins	rail	decoupling
17, 53, 96, 132	2V5	C8-C11, 100 nF each
9, 16, 36, 43, 70, 86, 100, 122, 137	3V3	C12-C20, 100 nF each
8, 15, 21, 32, 42, 65, 75, 85, 87, 101, 123, 129, 131, 138	GND	one via each into the In1 plane
109, 144	—	not connected

5. Connector pinouts

J1 — JTAG (PinHeader_1x10_P2.54mm_Vertical)

pin	net		pin	net
1	3V3		7	GND
2	TD0		8	TCK
3	TDI		9	DONE
4	PROGRAMN		10	INITN
6	TMS			``

J2 — JTAG-1.27 (PinHeader_2x05_P1.27mm_Vertical)

pin	net		pin	net
1	3V3		5	TCK
2	TMS		6	GND
3	TDI		10	GND
4	TD0			``

J3 — EXP-A (PinHeader_2x10_P2.54mm_Vertical)

pin	net		pin	net
1	3V3		11	spi0_cs_n
2	GND		12	spi0_sck
3	A8		13	spi0_mosi
4	A9		14	spi0_miso
5	A10		15	GND
6	A11		16	GND
7	A12		17	5V_USB
8	A13		18	5V_USB
9	A14		19	GND
10	A15		20	GND

J4 — EXP-B (PinHeader_2x10_P2.54mm_Vertical)

pin	net	pin	net
1	3V3	11	B8
2	GND	12	B9
3	B0	13	B10
4	B1	14	B11
5	B2	15	B12
6	B3	16	B13
7	B4	17	B14
8	B5	18	B15
9	B6	19	GND
10	B7	20	5V_USB

J5 — EXP-C (PinHeader_1x10_P2.54mm_Vertical)

pin	net	pin	net
1	A0	6	A5
2	A1	7	A6
3	A2	8	A7
4	A3	9	GND
5	A4	10	3V3

J6 — MOTOR (PinHeader_1x06_P2.54mm_Vertical)

pin	net	pin	net
1	VM_IN	4	motor_dir1
2	GND	5	motor_dir2
3	pwm_out	6	motor_fault_n

J8 — USB-C (USB_C_Receptacle_HRO_TYPE-C-31-M-12)

⚠ **Confirm the footprint:** USB-C 16-pin receptacle: this is the HRO TYPE-C-31-M-12 land pattern; a different part needs its own footprint (micro-USB drops R36/R37)

pin	net	pin	net
A1	GND	B6	USB_DP
A4	USB_VBUS	B7	USB_DN
A5	CC1	B9	USB_VBUS
A6	USB_DP	S1	GND
A7	USB_DN	S2	GND
A9	USB_VBUS	S3	GND
B1	GND	S4	GND
B4	USB_VBUS	A12	GND
B5	CC2	B12	GND

J9 — 7-12 V (BarrelJack_CUI_PJ-102AH_Horizontal)

⚠ **Confirm the footprint:** 5.5/2.1 mm barrel jack: confirm against the part you buy

pin	net	pin	net
1	J9_VIN	2	GND

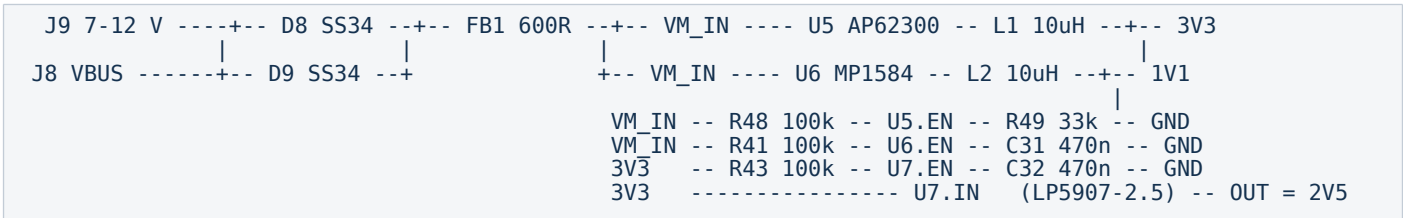
J10 — CONSOLE (PinHeader_1x04_P2.54mm_Vertical)

pin	net	pin	net
1	3V3	3	J10_RX
2	J10_TX	4	GND

JP1 — DTR ENABLE (PinHeader_1x02_P2.54mm_Vertical)

pin	net	pin	net
1	U4_DTR	2	DTR_F

6. Power tree and sequencing



Rail	Source	Set by	Rises	Bulk
3V3	U5 AP62300 (sync buck, 750 kHz)	R46 33k / R47 10k on FB → 3.28 V	first, from VM_IN	C24 220 μF + C40/C41 22 μF + C25 10 μF
1V1	U6 MP1584 (buck, 900 kHz via R40)	R38 12.4k / R39 33k on FB → 1.10 V	≈13 ms after VM_IN	C21 100 μF (tantalum) + 100 nF
2V5	U7 LP5907 from 3V3	RC on EN: R43 100k + C32 470n → ≈21 ms	≈21 ms after 3V3	C42 10 μF (low-ESR, mandatory) + C22 22 μF

Two things that break this board if they are wrong, both from `PCB_COMPONENTS.md` §10:

- **U5 pin 5 (EN) is a 6 V pin** — it is driven by the R48/R49 divider (3.0 V at 12 V in), never by VM_IN directly.
- **U6 pin 3 (COMP)** needs its RC network (R45 / C35, already placed as components) — the MP1584 is not stable without it.

Current plan (`BOARD.md` §4.2): 3V3 ≈ 600 mA, 1V1 ≈ 150 mA, 2V5 ≈ 50 mA. Copper: 0.8 mm for the rails, 1.5 mm for VM_IN, pours for everything local. 266 vias are already placed, including one on every GND pad that could take one.

7. Rail delivery — what is already copper, and what the router owes

The board file ships with the rail pours on `In2.Cu` and with the vias below already placed, each one wrapped in a 0.2 mm stub track from its pad (that is how an SMD pad reaches an inner layer).

net	copper
3V3	whole-layer pour on In2.Cu, priority 0

net	copper
2V5	patch pour: LP5907 group + the C8-C11 band above U1
1V1	patch pour: MP1584 group + the strip down the left of U1 (C2-C7)
GND	solid plane on In1.Cu + pour on B.Cu

Vias already placed: 1V1 11, 2V5 8, 3V3 52, GND 195. Of those, 110 are GND stitching round the board edge, and the rest are escape vias on pads that sit over their own rail's pour.

These pads are over a different rail's pour, so a via would short two rails. Route each one with a 0.25 mm track to its own patch (or move it in the placement and regenerate):

- U1.130 (1V1) escapes into the 3V3 pour - route it out by hand, do not via here
- U1.132 (2V5) escapes into the 3V3 pour - route it out by hand, do not via here
- U1.17 (2V5) escapes into the 3V3 pour - route it out by hand, do not via here
- U1.38 (1V1) escapes into the 3V3 pour - route it out by hand, do not via here
- U1.66 (1V1) escapes into the 3V3 pour - route it out by hand, do not via here
- U1.83 (1V1) escapes into the 3V3 pour - route it out by hand, do not via here
- U1.96 (2V5) escapes into the 3V3 pour - route it out by hand, do not via here

Everything else is a normal ratsnest connection: run `make pcb` then open the board, press **B** to fill the zones, and route.

8. The connection sheets, drawn

The same thing as §3-§7 but as pictures — `hardware/sv16_board/pcb_connection_sheets.png` holds all six at full size. If you are drawing the schematic, these are the sheets to copy.

Sheet 1 — power into the FPGA

```
J9 7-12 V --+-- D8 SS34 ---+-- FB1 ---+-- VM_IN ---+-- U5 AP62300 -- L1 10uH -- 3V3
      |               |               |               |
J8 VBUS ----+-- D9 SS34 ---+          C23 470u  +-- U6 MP1584EN -- L2 10uH -- 1V1
      |               |               |               |
      C39 10u          |               |               |
                      +-- D11 SS34 (catch) C21 100u
U7 LP5907-2.5: 3V3 -- IN ; EN <-- R43 100k -- 3V3 , C32 470n -> GND (~21 ms)
                OUT -- 2V5 -- C42 10u + C22 22u

U1 supplies:  1V1 -> pins 20 29 38 66 83 130  + C2-C7  100n each
               2V5 -> pins 17 53 96 132      + C8-C11 100n each
               3V3 -> pins  9 16 36 43 70 86 100 122 137 + C12-C20 100n each
               GND -> pins  8 15 21 32 42 65 75 85 87 101 123 129 131 138
               NC  -> pins 109, 144
```

Sheet 2 — configuration, JTAG, reset, status

```
U1.57 PROGRAMN ---+-- R2 4.7k -- 3V3      U1.63 TCK -- J1.8 -- J2.5
               +-+ SW2 -- GND             U1.64 TMS -- J1.6 -- J2.2
               +-+ J1.4                   U1.61 TDI -- J1.3 -- J2.3
               +-+ Q1 drain                U1.60 TD0 -- J1.2 -- J2.4
               Q1 source -- GND            J1.7 / J2.6 /

J2.10 -- GND

               Q1 gate -- R25 4.7k -- JP1 -- U4 DTR#
               Q1 gate -- R26 10k -- GND

U1.55 INITN ---+-- R3 4.7k -- 3V3      U1.56 DONE ---+-- R4 4.7k -- 3V3
               +-+ J1.10                  +-+ J1.9
               +-+ R31 1k -- Q4 base        +-+ R32 10k -- Q2 base
               Q4 (PNP) emitter -- 3V3      Q2 (NPN) emitter -- GND
               Q4 collector -- D6 -- R34 1k -- GND  Q2 collector -- D1 -- R33 1k -- 3V3

mode straps:  CFG_0 (62) -- R6 1k -- GND  CFG_1 (59) -- R5 4.7k -- 3V3  CFG_2 (58) -- R7 1k -- GND
reset:        ext_rst_n (134) -- SW1 -- GND , R1 10k -- 3V3 , C1 100n -- GND
```

Sheet 3 — the two flashes

```
U2 configuration flash (Master SPI, driven by the configuration logic)
U1.54 CCLK -- R9 100R --> U2.6 CLK          U1.54 -- R8 1k -- 3V3
U1.47 MOSI -- R10 100R -> U2.5 DI           U2.1 /CS -- R13 10k -- 3V3
U1.46 MISO -- R11 100R -> U2.2 D0          U2.3 /WP -- R14 10k -- 3V3
U1.49 CSSPIN - R12 100R -> U2.1 /CS        U2.7 /HOLD-- R15 10k -- 3V3
                                           U2.8 VCC = 3V3 + C28 100n ; U2.4 GND

U3 application flash (driven by the soft core, sv16_flash_ctrl at 0xF0A0)
U1.110 flash_sck -----> U3.6 CLK          U3.1 /CS-- R16 10k -- 3V3
U1.111 flash_cs_n -----> U3.1 /CS        U3.3 /WP -- R17 10k -- 3V3
U1.112 flash_mosi -----> U3.5 DI          U3.7 /
HOLD- R18 10k -- 3V3
U1.113 flash_miso <----- U3.2 D0          U3.8 VCC = 3V3 + C29 100n
```

Sheet 4 — clock, reset, console

```
Y1 25.000 MHz ACTIVE XO          U4 CH340G (SOP-16)          J8 USB-C
1 EN -- 3V3                      2 TXD -- R21 0R -- U1.73          A4 B4 A9 B9 -- VBUS
2 GND -- GND                     3 RXD -- R23 0R -- U1.74          A6/B6 D+ -- U8 -- D+/D- U4.5
3 OUT -- U1.133 clk_25m          4 V3 -- 3V3 + C34 100n          A7/B7 D- -- U8 -- D-/D+ U4.6
4 VCC -- 3V3 + C26 100n          7 XI / 8 XO -- Y2 12 MHz + C36/C37  A5 CC1 -- R36 5.1k -- GND
                                   13 DTR# -- JP1 -- R25 -- Q1 gate      B5 CC2 -- R37 5.1k -- GND
                                   16 VCC -- 3V3 + C27 100n          S1-S4 -- GND
                                   5 UD+ / 6 UD- -- U8

SW1/R1/C1 on U1.134

J10 console alternative: fit R22/R24 and REMOVE R21/
R23.
```

Sheet 5 — expansion and motor

```
J3 EXP-A 2x10    1 3V3    2 GND    3-10 A8..A15    11 spi0_cs_n 12 spi0_sck
                  13 spi0_mosi 14 spi0_miso 15,16 GND 17,18 5V(VBUS) 19,20 GND
J4 EXP-B 2x10    1 3V3    2 GND    3-18 B0..B15    19 GND    20 5V(VBUS)
J5 EXP-C 1x10    1-8 A0..A7    9 GND    10 3V3    (pins 2/3/5/7 also reach U2: INPUTS ONLY)
J6 MOTOR 1x6     1 VM_IN  2 GND    3 pwm_out  4 motor_dir1 5 motor_dir2 6 motor_fault_n
J7 SPARE 2x25    footprint only, not fitted (46 spare I/
0 + 4 GND)
LEDs             led[0..3] (39 40 41 44) -- R27-R30 470R -- D2-D5 -- 3V3 (active low)
```

Sheet 6 — build order and first power-up

```
1 power section only: U5 U6 U7 L1 L2 D8 D9 D11 FB1 C23 C24 C39 C40 C41 J9
2 bench supply 12 V, current limit 200 mA, through J9
3 TP1 = 3.28 V TP3 = 1.10 V TP2 = 2.50 V (≈21 ms after 3V3) TP4 = 0 V
4 wrong rail? -> R46/R47 (3V3) · R38/R39 (1V1) · R43/
C32 (2V5) · U5 EN ≈3 V, never 12 V
5 fit the rest, U1 (FPGA) last
6 power again, re-measure all four test points
7 USB: the CH340G must appear as a serial port
8 configure over J1 (openFPGAloader), then write the config flash
9 DONE (TP5) high, monitor answers '?' at 115200 8-N-1
10 DONE never rises -> scope clk_25m at Y1 pin 3, then check PROGRAMN, CCLK, U2
```

9. Before you route — the twelve checks

#	Check	Why
1	Open the project (sv16_board.kicad_pro), not the bare board	it carries the five net classes and their widths
2	Tools → Update Footprints from Library	the board ships with generated pads: correct numbers, positions and nets, approximate sizes
3	141 footprints, 121 nets, 472 pads with a net, 291 vias, 163 stubs	the sanity check; <i>Inspect → Net Inspector</i>
4	Y1 within 10 mm of U1 pin 133	a long clock trace picks up the switchers

#	Check	Why
5	U2 within 15 mm of U1 pins 54/46/47/49, with R9–R12 at the FPGA end	this is the configuration path the FPGA needs at power-on
6	The switchers (U5, U6, L1, L2, C38–C41) away from Y1, U2 and U4	that is where the EMI comes from
7	U8 between J8 and U4, J8 flush with the edge, R36/R37 within 10 mm of J8	the ESD device protects the chip only if it is on the connector side
8	B — fill all zones — then look for islands on In1.Cu	an unfilled plane is not a plane; islands are DRC warnings worth fixing
9	Check the three In2 pours do not leave a rail pad stranded	§7 lists the pads that are over the wrong pour
10	DRC with refill zones ticked; fix copper errors, ignore silkscreen ones for now	every copper error is a spinner or a short
11	Add a third via size 0.45 / 0.2 mm for the QFP fan-out	the default 0.8/0.4 via will not fit between 0.5 mm pads
12	Route in this order: power → switcher loops → QFP fan-out → USB pair → clocks → the rest	it is the order that keeps the inner layers free for the pours

Two numbers to keep in front of you while routing: **0.2 mm** track and clearance for the FPGA fan-out, and **0.8 mm** for anything carrying a rail.

10. Bring-up: measure, in this order

Step	Do	Expected
1	Fit only the power section and J9	U5, U6, U7, L1, L2, D8, D9, D11, FB1, C23, C24, C39–C42
2	Bench supply at 12 V, 200 mA limit	a few tens of mA — if the limit trips, find the short before reapplying
3	TP1	3.28 V (R46/R47)
4	TP3	1.10 V (R38/R39, R45/C35)
5	TP2	2.50 V , appearing ≈21 ms after 3V3 (R43/C32)
6	TP4	0 V
7	Fit the rest, U1 last	flashes before the FPGA, headers and connectors any time
8	Power again, re-measure TP1–TP3	a sagging 3V3 with U1 fitted = a short under the QFP or a missing decoupling cap
9	Plug in USB	the CH340G appears as a serial port (dmesg, or Device Manager with the WCH driver)
10	Configure over J1 (make prog), then make prog-flash	DONE (TP5) goes high when the FPGA is configured
11	Open the console at 115200 8-N-1, send ?	the monitor prints its help

Step	Do	Expected
12	If DONE never rises	scope <code>clk_25m</code> at Y1 pin 3 → then PROGRAMN (57), CCLK (54) and U2

11. Still to buy (the parts the documents say are missing from the cart)

Ref	What	Why the board does not work without it
R47	10 kohm 1 %	AP62300 feedback bottom leg - the rail will not regulate without it
C39	10 uF X5R 25 V	AP62300 input ceramic
C40, C41	22 uF X5R 25 V	AP62300 output ceramics
C42	10 uF low-ESR X5R	LP5907 output - it is only stable into a low-ESR capacitor
C25, C30	10 uF	3V3 local bulk and the LP5907 input
Y1	active 25 MHz 3.3 V XO, 4-pin	the HC49/US crystal in the cart cannot clock the FPGA
Y2	12 MHz crystal + 22 pF x2	only if U4 is the CH340G; the C variant needs neither

And one decision that is yours: the **470 µF input bulk (C23) in the cart is a 16 V part on a 12 V rail** — 1.33× derating where 35 V was specified. Fine on a bench supply that never exceeds 12 V; use 25–35 V if you can still change it.

12. Regenerating this file

```
make pcb          # schematic + board file + bitmaps + BOM/CPL + this document
make schematic    # just the KiCad schematic (sv16_board.kicad_sch)
make pcb-check    # parse the schematic and the board back, then re-check both
python3 scripts/sv16_pcb_connections.py --check # is this document stale?
```

Revision 1.0 — generated from commit data in scripts/

sv16_board_kicad.py (netlist), scripts/

sv16_pcb_copper.py (copper plan) and constraints/

ecp5_144tqfp.lpf (pin map).